RepairDesk Location Features — Implementation Task List (Google Maps)

Related: Proposed app wireframes (screen numbers below refer to that document)
Map provider: Google Maps Platform, India pricing
Status: Plan for review — 10 October 2026

Effort: S = up to 1 day · M = 2–3 days · L = about 1 week (one developer). **Owner:** *You* = business owner / account holder · *Dev* = development.

The phases follow the rollout in the wireframes document. Each phase can ship on its own, and later phases build on the earlier ones.

Phase 0 — Google Maps account and keys

This phase needs the business owner. Google requires the account holder to accept its terms and add a payment method, so it can't be done on your behalf.

#	Task	Owner	Effort
0.1	Create (or choose) a **company-owned Google account** for Google Cloud, not a personal Gmail, so access survives staff changes	You	S
0.2	Open Google Cloud Console and create a **billing account** with an **Indian billing address** and GSTIN, so India pricing and its larger free limits apply. Add a card.	You	S
0.3	Create two projects: `repairdesk-dev` and `repairdesk-prod`	You or Dev	S
0.4	Enable APIs in both projects: **Maps SDK for Android**, **Maps JavaScript API**, **Geocoding API**, **Routes API**. Optional later: **Places API (New)** for address autocomplete	You or Dev	S
0.5	Create three restricted API keys per project: an **Android key** (restricted to app `com.acetelecom.repairdesk` and its signing SHA-1 fingerprints), a **web key** (restricted to the admin site domain), and a **server key** (restricted to Geocoding and Routes, used only by the backend)	Dev	S
0.6	Set **daily quotas**: Geocoding, Routes and Maps JavaScript at 2,250 requests per day each. Add a **budget alert** of ₹500 per month with emails at 50%, 90% and 100%	Dev	S
0.7	Give the developer access: add their Google account to both projects (role **Editor** on dev, **API Keys Admin** + **Viewer** on prod). Never send keys by chat or email, and never commit them to the repo	You	S
0.8	Store the server key as a secret in the Azure Container App (and the dev environment); store the Android key in the CI / build machine, not in git	Dev	S

Phase 1 — Foundations

#	Task	Area	Owner	Effort
1.1	**Service address and map pin.** The backend currently stores no customer service address (only the shop address in settings). Add address line, landmark, city, PIN code, latitude, longitude and pin source to requests and jobs (Flyway migration `V12`)	Backend	Dev	M
1.2	**Shop location and settings.** Add shop latitude, longitude and geofence radius to business settings, plus a location-settings record: feature switches, ping interval, working-hours end, far-from-address threshold, allowance rates, retention days	Backend	Dev	M

#	Task	Area	Owner	Effort
1.3	**Maps service in the backend.** One class that calls Geocoding and Routes with the server key, caches results (the same address is never geocoded twice), counts calls per day, and falls back to straight-line distance when the daily limit is reached	Backend	Dev	M
1.4	Admin API: `GET/PUT /settings/location`	Backend	Dev	S
1.5	Flutter packages: `google_maps_flutter`, `geolocator`, `permission_handler`, `url_launcher`. Add location and notification permissions to `AndroidManifest.xml`; inject the Android key into the manifest at build time (Gradle placeholder), not in source	App	Dev	S
1.6	**Permission onboarding**: Screen 1A explanation, Android prompts (1B), notification prompt, and the three banners (location off, approximate, only this time). Brand-specific battery help for Xiaomi, Realme, Vivo, Oppo	App	Dev	M
1.7	Admin site: add `@angular/google-maps`; **address + pin** on the Staff "create request" form and request detail	Admin web	Dev	M
1.8	Admin site: **Location settings** page (Screen 8) with the shop pin, radius and feature switches	Admin web	Dev	M

Phase 2 — Geotagged check-in and check-out (feature 1)

#	Task	Area	Owner	Effort
2.1	`job_visits` table: job, technician, type (On my way / Check-in / Check-out), latitude, longitude, accuracy, time, distance from customer pin, low-accuracy flag	Backend	Dev	S
2.2	Accept an optional location with **Start Repair** and **Mark Complete**; calculate distance from the pin and the far-from-address flag	Backend	Dev	M
2.3	App: capture the best location fix within 20 seconds in the Start Repair and Mark Complete confirmations; never block the job if GPS is weak	App	Dev	M
2.4	App: **Visit** section on Job Detail with map, distance, and **Navigate** (opens the Google Maps app); **Visit record** after completion (Screen 3)	App	Dev	M
2.5	Save the customer pin from the first check-in when the job has none	Backend + App	Dev	S
2.6	Staff and Admin: Visit record, far-from-address flag and "View on map" on job detail (Screen 6)	App + Admin web	Dev	M
2.7	Tests: distance and flag unit tests; emulator tests with simulated locations	All	Dev	M

Phase 3 — Attendance by geofence (feature 2)

#	Task	Area	Owner	Effort
3.1	`attendance_days` table and **Start day / End day** endpoints; record whether each was inside the shop radius	Backend	Dev	M
3.2	Scheduled job: **auto-end** the day at the working-hours end time	Backend	Dev	S
3.3	App home: duty card with inside / outside shop area and the disabled state (Screen 2)	App	Dev	M

#	Task	Area	Owner	Effort
3.4	**Attendance & travel report** for Staff and Admin, with Excel (CSV) export (Screen 6)	App + Admin web	Dev	M

Phase 4 — On-duty tracking (feature 3)

#	Task	Area	Owner	Effort
4.1	App: **background tracking** between Start day and End day — Android foreground service with the "sharing location while on duty" notification, background-location permission via Settings, ping every N minutes from settings	App	Dev	L
4.2	App: **offline queue** — store points on the phone and upload in batches when the network returns	App	Dev	M
4.3	`location_points` table (indexed by technician and time), batch upload endpoint, and a nightly job that deletes points older than the retention period	Backend	Dev	M
4.4	**Live technician map** for Staff (app) and Admin (web), with a warning after 15 minutes without a ping (Screen 5)	App + Admin web	Dev	L
4.5	Optional: sort technicians by distance in the Assign dialog	All	Dev	S
4.6	**Field test on budget phones** (Redmi, Realme, Vivo, Samsung M/A series): gaps, battery use per day, behaviour after reboot	QA	Dev + You	M

Phase 5 — Distance and travel allowance (feature 4)

#	Task	Area	Owner	Effort
5.1	Daily distance from on-duty points: filter GPS jumps and jitter, split into legs at job check-ins	Backend	Dev	M
5.2	`travel_claims` table; submit, approve (single and bulk) and monthly export	Backend	Dev	M
5.3	App: **Travel & allowance** screen and Submit claim (Screen 4)	App	Dev	M
5.4	Staff / Admin: approve claims in the report	App + Admin web	Dev	S
5.5	Confirm allowance rates and approval rules (two-wheeler / four-wheeler per km)	—	You	S

Phase 6 — Customer live ETA (feature 5)

#	Task	Area	Owner	Effort
6.1	**On my way** endpoint: start a tracking session with a private link; ETA from the Routes API every 2 minutes (cached); stop automatically at check-in	Backend	Dev	M
6.2	Push notification to the customer app; SMS with the tracking link	Backend	Dev	M
6.3	**SMS setup**: DLT registration of the sender ID and message template with an SMS provider (takes 1–2 weeks of lead time)	—	You	S
6.4	Customer app tracking screen (Screen 7) and a lightweight public tracking web page for customers without the app	App + Web	Dev	L

Phase 7 — Privacy, release and documentation

#	Task	Area	Owner	Effort
7.1	Privacy notice for technicians (DPDP Act 2023): what is collected, why, duty hours only, retention. Record each technician's acceptance with date and version	App + Backend	Dev + You	M
7.2	Technicians can view and download their own location history from Account	App	Dev	S
7.3	If publishing on the Play Store: background-location declaration, demo video, Data safety form	—	Dev + You	S
7.4	Update the Technician, Staff and Admin manuals and regenerate the PDFs	Docs	Dev	M
7.5	Monitoring: Google Cloud usage dashboard, budget alerts, backend count of map calls per day	Ops	Dev	S

Decisions needed before Phase 1

Question	Proposed default
Shop geofence radius	150 m
Can technicians start the day outside the shop?	No; Admin can enable per shop
Tracking interval	5 minutes
Working hours / auto end of day	9:00 AM – 7:00 PM
Allowance rates	₹5 / km two-wheeler, ₹10 / km four-wheeler
Location history retention	90 days
Technician phones	Personal phones (prompts) or company phones (pre-granted via `adb install -g` or device management)
Distribution	APK only, or Play Store (adds the background-location review)